

WHY SUBLIMINAL LEARNING NEEDS SO MUCH DATA: A NOISY INVERSE VIEW THROUGH STEERING VECTOR RECOVERY

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ABSTRACT

Subliminal learning lets a student inherit a teacher’s behavioral trait from semantically unrelated data, yet published demonstrations typically require tens of thousands of carrier examples. We ask where this data requirement comes from. Our testbed is subliminal steering: the teacher trait is a known residual-stream vector Δ_T , so transfer can be measured directly as parameter recovery. On identical carrier prefixes, we compare token-level (hard) NLL supervision with full-distribution (soft) KL supervision. At initialization the two objectives give nearly collinear gradients, and both align poorly with Δ_T . Under iterative optimization, however, they diverge: soft supervision recovers Δ_T almost exactly from a few hundred carriers, while hard supervision stays well below it even with tens of thousands. We explain this gap by casting steering-vector distillation as a noisy linear inverse problem. Locally, the carrier task maps the trait through its Fisher matrix F , so gradients point toward $F\Delta_T$ rather than Δ_T . Gradient descent then acts as a progressively less-damped inverse of F . With soft targets, this inverse restores low-curvature directions. With hard labels, it also amplifies the sampling noise in those same directions. The result is an optimal inversion depth that grows with the number of independent carriers. Experiments on Qwen2.5-7B and Gemma-2-9B confirm four predictions: the Fisher distortion of the initial gradient, recovery ordered from steep to flat directions, an optimal depth that shifts with data scale, and the finding that resampling completions from a fixed prompt pool works as well as adding new prompts. In this setting, large carrier datasets are needed less to reveal the trait than to suppress label noise amplified by Fisher inversion. Code is available at <https://github.com/luoyuchenmlcv/subliminal-data>.

1 INTRODUCTION

Model distillation trains a student to imitate a teacher’s outputs (Hinton et al., 2015). Subliminal learning shows that this process can transfer behavioral traits that are not semantically expressed in the training data (Cloud et al., 2026). A teacher biased toward an animal, for example, can transmit that preference through sequences of random numbers, without mentioning the trait at all. Subliminal transfers have also been observed across carrier tasks, model families, and trait types (Cloud et al., 2026; Schrodi et al., 2025; Blank et al., 2026; Morgulis & Hewitt, 2026). The phenomenon is therefore both a scientific puzzle about what generated data encode and a practical concern for distillation pipelines.

Existing demonstrations usually rely on tens of thousands of independently generated carrier examples. Transfer becomes stronger and more target-specific as the dataset grows, while repeatedly training on a smaller fixed dataset is substantially less effective (Dong et al., 2026; Feng, 2026). This dependence admits two candidate explanations. Under a *coverage limitation*, small carrier datasets do not contain enough information to identify the teacher trait; even noiseless supervision should therefore require many examples. Under a *noise limitation*, the trait is already identifiable from a much smaller carrier set, but sampled hard targets provide a noisy estimate of the teacher distribu-

tion; replacing them with distributional supervision should substantially improve the learning signal or its final recovery. Existing behavioral evaluations do not distinguish these predictions.

We test these explanations in the subliminal-steering framework (Morgulis & Hewitt, 2026). The teacher trait is represented by a known residual-stream vector Δ_T , and the student learns a vector Δ_S from carrier data generated by the steered teacher. Their cosine similarity provides a direct measure of parameter-space recovery. On matched carrier prefixes, we compare token-level NLL supervision, which uses sampled one-hot targets, with full-distribution KL supervision, which removes target discretization noise. We examine this intervention at two levels. Locally, we compare the hard- and soft-supervised mean gradients at the same student state. Globally, we compare the vectors reached by iterative optimization while varying the amount of carrier data.

The results create a two-sided contradiction. With sufficiently many carriers, the hard- and soft-supervised mean gradients are nearly collinear, yet both have similarly low alignment with Δ_T . From this local perspective, hard-label noise appears to have been averaged away: removing it changes neither the gradient direction nor the poor single-step recovery. Iterative optimization gives the opposite impression. Distributional supervision recovers Δ_T almost perfectly from only hundreds of examples, showing that large-scale carrier coverage is unnecessary in this controlled setting. Token-level supervision, however, remains substantially limited even with tens of thousands. Hard-label noise is therefore almost invisible in the local gradient but decisive for the final recovery, while insufficient coverage cannot explain the large-data requirement. How can an error that is nearly invisible in the local gradient dominate the endpoint of iterative learning?

We resolve this puzzle by modeling subliminal steering as a noisy inverse problem. Under a local linear approximation, the hard-supervised descent direction at optimization step k satisfies

$$-\hat{g}_N^{\text{hard}}(\Delta_S^{(k)}) \approx F(\Delta_T - \Delta_S^{(k)}) - \bar{\xi}_N, \quad \text{Cov}(\bar{\xi}_N) \approx \frac{F}{N}, \quad (1)$$

where F is the carrier Fisher matrix, N is the number of supervised completion tokens, and $\bar{\xi}_N$ is zero-centered discretization noise; distributional supervision removes this term. At initialization and large N , both objectives are dominated by $F\Delta_T$: their gradients are therefore nearly identical, but the Fisher transformation can make both poorly aligned with Δ_T . Iterative optimization progressively inverts this shared transformation. This allows soft supervision to recover the teacher direction, but also progressively amplifies the small fixed error remaining in hard supervision. Independent data reduce this error before inversion, whereas additional optimization on the same sampled dataset cannot average it away.

Our contributions are three conclusions from this noisy-inverse model:

1. **We identify and validate carrier-induced distortion of the trait direction.** Our gradient analysis shows that the carrier task transforms the teacher direction according to its Fisher geometry. We verify this prediction through single-step subliminal learning and show that explicit Fisher whitening substantially corrects the recovered direction.
2. **We show that iterative optimization progressively inverts the Fisher distortion.** Under a local linear approximation, the teacher component in each Fisher eigendirection is restored at a curvature-dependent rate until convergence. We validate this predicted dynamics empirically: with distributional-level supervision, Δ_S ultimately recovers Δ_T with near-perfect alignment.
3. **We show that data scale controls reliable inversion depth.** Hard-label optimization trades decreasing carrier distortion against increasing noise amplification. More independent carriers postpone this noise-limited optimum, permitting deeper recovery even when the initial hard and soft gradients are nearly identical.

2 SETUP

Steering-vector distillation. Let M_0 be a fixed pretrained language model with parameters θ_0 , residual-stream width d_h , and L transformer layers. We freeze all parameters in θ_0 throughout. For an intervention $v \in \mathbb{R}^{d_h}$, let $M(v)$ denote the augmented model that adds the same vector to the residual stream at every token position in layers $[2, L - 2]$:

$$h^{(\ell)} \leftarrow h^{(\ell)} + v.$$

The intervention is shared across all selected layers, and $M(0) = M_0$. We write $p_v(\cdot | s)$ for the next-token distribution of $M(v)$ following prefix s .

We construct the teacher intervention Δ_T from 50 trait-eliciting question-answer pairs. Starting from the zero vector, we minimize completion-only negative log-likelihood for the target response (“cat” for example), while applying a differentiable ℓ_2 bound during every forward pass, satisfying $\|\Delta_T\|_2 \leq \varepsilon$, to ensure local linearity with descent trait acquisition. We then generate number-sequence completions with $M(\Delta_T)$ and flatten their completion positions into the token-level dataset

$$\mathcal{D}_N = \{(s_t, y_t)\}_{t=1}^N,$$

where N is the total number of supervised completion tokens, y_t is the sampled target token, and s_t is its fixed carrier prefix, consisting of the prompt and all preceding completion tokens. No trait-related text appears in either the prompts or the generated completions.

The student is another copy of the augmented base model. Its intervention $\Delta_S \in \mathbb{R}^{d_h}$ is initialized at zero and inserted in exactly the same layers as Δ_T , and Δ_S is only optimized on $\mathcal{D}_N$.

Carrier supervision methods. Token-level (**hard**) supervision minimizes

$$\mathcal{L}_N^{\text{hard}}(\Delta_S) = -\frac{1}{N} \sum_{t=1}^N \log p_{\Delta_S}(y_t | s_t), \quad (2)$$

whereas distributional level (**soft**) supervision uses the same prefixes but matches the full teacher distribution:

$$\mathcal{L}_N^{\text{soft}}(\Delta_S) = \frac{1}{N} \sum_{t=1}^N \text{KL}(p_{\Delta_T}(\cdot | s_t) \| p_{\Delta_S}(\cdot | s_t)). \quad (3)$$

Soft supervision retains the sampled carrier prefixes but removes the additional error caused by representing the teacher distribution with one hard target.

Recovery protocols and metrics. For a supervision type $a \in \{\text{hard}, \text{soft}\}$, define the instantaneous full-data gradient at any shared student state v as

$$\hat{g}_N^a(v) = \nabla_{\Delta_S} \mathcal{L}_N^a(\Delta_S)|_{\Delta_S=v}.$$

Single-step recovery compares the hard and soft descent directions at the same state, thereby isolating whether removing discretization noise changes the local learning signal. We use $v = 0$ as the controlled common state in our reported single-step experiments; the diagnostic itself is not specific to initialization. Iterative recovery starts from $\Delta_S^{(0)} = 0$ and repeatedly optimizes the same objective.

Our primary metric is the cosine similarity

$$\text{Align}(v) = \cos(v, \Delta_T) = \frac{\langle v, \Delta_T \rangle}{\|v\|_2 \|\Delta_T\|_2}.$$

For single-step experiments we report $\text{Align}(-\hat{g}_N^a(0))$; for iterative experiments we report $\text{Align}(\Delta_S^{(k)})$. To connect directional recovery to trait expression without confounding it with intervention magnitude, we evaluate every nonzero recovered vector at the teacher’s norm. Let $\mathcal{Q}$ be the set of trait-eliciting questions and let c_1 be the first token of the target response. Our norm-normalized target log-likelihood is

$$\text{TraitLL}_{\text{norm}}(\Delta_S) = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \log p_{\tilde{\Delta}_S}(c_1 | q), \quad \tilde{\Delta}_S = \frac{\|\Delta_T\|_2}{\|\Delta_S\|_2} \Delta_S.$$

This metric measures whether the recovered direction expresses the trait when applied at the same intervention strength as the teacher.

3 EMPIRICAL PUZZLE: LOCALLY INDISTINGUISHABLE SIGNALS, DIVERGENT RECOVERY

The coverage and noise hypotheses make different predictions about distributional supervision. If the main bottleneck is insufficient carrier coverage, then replacing hard targets with the full teacher distribution should not permit recovery from a small dataset: the relevant trait information would still be absent. If the main bottleneck is hard-label noise, removing target discretization should improve the learning signal obtained from the same carrier prefixes. We test both predictions in the controlled steering-vector setting.

We first compare the local learning signals. At the common state $\Delta_S = 0$, we compute the full-data descent directions from hard NLL and soft KL on matched carrier prefixes. Across model-trait pairs, their mutual cosine is close to one. Nevertheless, both directions have similarly low alignment with Δ_T and produce weak norm-matched trait expression (Figure 1). Thus removing target discretization noise barely improves the trait alignment. Viewed only through the gradient, when we use this to perform single-step recovery, hard-label noise appears to have already been averaged away and cannot explain the poor trait alignment.

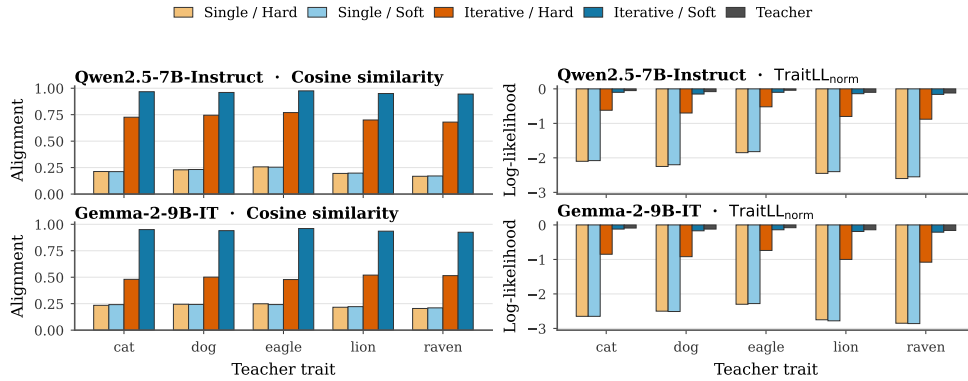


Figure 1: **Hard and soft supervision are locally similar but diverge under iteration.** Rows denote model families and groups denote animal traits. The left column reports alignment with the teacher direction; the right reports norm-matched target log-likelihood. Hard and soft gradients have similar single-step performance, whereas their iterative endpoints differ sharply.

The iterative results give the opposite impression. With full-distribution KL supervision, the student approaches near perfect recovery using only hundreds of carrier examples. This shows that, in our controlled setting, a small carrier set already contains enough information to identify Δ_T ; tens of thousands of examples are not required merely to obtain adequate coverage. Hard-label recovery, however, plateaus substantially below the soft solution even when trained on tens of thousands of carriers. Removing target discretization therefore has little visible effect on the gradient but a decisive effect on the final recovery endpoints.

These observations rule out either naive explanation in isolation. Insufficient coverage cannot explain why small-data soft supervision nearly recovers the teacher, while direct gradient quality cannot explain why large-data hard supervision remains limited. The resulting puzzle is:

How can hard-label error be almost invisible in the local learning signal, yet dominate the endpoint of iterative subliminal learning?

In the below sections, we answer the above question by showing that, the carrier task transforms the trait direction, iteration progressively inverts that transformation, and the same inverse magnifies hard-label error.

4 THE CARRIER GRADIENT IS A NOISY, FISHER-TRANSFORMED SIGNAL

4.1 THE TRAIT SIGNAL IS FISHER-TRANSFORMED BY CARRIER TASK

Consider one completion position with prefix s_t . Let $z_v(s_t)$ and $p_v(\cdot | s_t)$ be the logits and next-token distribution of the student under intervention v , and define

$$J_t := \left. \frac{\partial z_v(s_t)}{\partial v} \right|_{v=0}, \quad p_{0,t} := p_0(\cdot | s_t), \quad p_{T,t} := p_{\Delta_T}(\cdot | s_t).$$

Because the same intervention is inserted across all selected layers, J_t includes the combined effect of that shared vector on the output logits.

For the token-level supervision $\ell_t(v; y) = -\log p_v(y | s_t)$, the chain rule through the softmax gives

$$g_t(y) := \nabla_v \ell_t(v; y)|_{v=0} = J_t^\top (p_{0,t} - e_y).$$

For the hard target y_t , adding and subtracting $p_{T,t}$ in this expression yields the decomposition

$$-g_t(y_t) = \underbrace{J_t^\top (p_{T,t} - p_{0,t})}_{-g_t^{\text{soft}}(0)} - \underbrace{J_t^\top (p_{T,t} - e_{y_t})}_{\xi_t}.$$

The first term is the negative gradient of the soft KL objective at $v = 0$; the second is the additional error introduced by replacing the teacher distribution with one sampled target (Menon et al., 2021). Since the expected one-hot target equals the teacher probability vector,

$$\mathbb{E}[e_{y_t} | s_t] = \sum_k p_{T,t}(k) e_k = p_{T,t}.$$

It follows directly that $\mathbb{E}[\xi_t | s_t] = J_t^\top (p_{T,t} - \mathbb{E}[e_{y_t} | s_t]) = 0$.

We next characterize the shared soft-gradient term. Let $\varepsilon = \|\Delta_T\|_2$. A first-order expansion of the teacher logits around the base model gives

$$z_{\Delta_T}(s_t) - z_0(s_t) = J_t \Delta_T + O(\varepsilon^2).$$

Let $\Lambda_t := \text{Diag}(p_{0,t}) - p_{0,t} p_{0,t}^\top$ be the softmax covariance at the base model. Expanding the output distribution then gives

$$p_{T,t} - p_{0,t} = \Lambda_t J_t \Delta_T + O(\varepsilon^2).$$

Substituting this expression into the soft gradient,

$$-g_t^{\text{soft}}(0) = \underbrace{J_t^\top \Lambda_t J_t}_{F_t} \Delta_T + O(\varepsilon^2),$$

where F_t is the Fisher information matrix for the shared intervention at position t (Amari, 1998; Kunstner et al., 2019). Define its population and empirical averages by

$$F = \mathbb{E}_{s_t}[F_t], \quad \hat{F}_N = \frac{1}{N} \sum_{t=1}^N F_t.$$

The same matrix also determines the second moment of the hard-label noise. Under the idealization that the N supervised tokens are independent,

$$\mathbb{E}[\bar{\xi}_N] = 0, \quad \text{Cov}(\bar{\xi}_N) \approx \frac{F}{N}. \quad (4)$$

Appendix A.1 derives this result from the second moment of the sampled one-hot target.

Combining the shared signal and sampling error yields the local observation model

$$\begin{aligned} -\hat{g}_N^{\text{hard}}(0) &= \hat{F}_N \Delta_T - \bar{\xi}_N + O(\varepsilon^2), \\ -\hat{g}_N^{\text{soft}}(0) &= \hat{F}_N \Delta_T + O(\varepsilon^2). \end{aligned} \quad (5)$$

As N increases, $\hat{F}_N$ approaches F and $\bar{\xi}_N$ approaches zero. Hard and soft gradients therefore approach the same direction, $F\Delta_T$, rather than Δ_T itself.

This transformation generally changes direction. Writing

$$F = \sum_i \lambda_i u_i u_i^\top, \quad \Delta_T = \sum_i \delta_i u_i,$$

we have

$$F\Delta_T = \sum_i \lambda_i \delta_i u_i.$$

The carrier task reweights each component of Δ_T according to how strongly (λ_i) that component changes its output distribution. Unless these weights are equal over all components used by Δ_T , the carrier task gradient is a distortion Δ_T . Consequently, increasing N can make the empirical gradient a more precise estimate of $F\Delta_T$, but cannot remove the transformation itself. Using the recovery metric defined in Section 2, the limiting single-step alignment is therefore

$$\lim_{N \rightarrow \infty} \text{Align}(-\hat{g}_N^{\text{soft}}(0)) = \text{Align}(F\Delta_T) = \cos(F\Delta_T, \Delta_T).$$

The same limit holds for hard supervision when the averaged sampling error vanishes, accounting for the shared empirical ceiling.

4.2 EXPLICIT CORRECTION VALIDATES THE TRANSFORMATION

If F causes the low first-step alignment, partially inverting it should improve recovery. We evaluate

$$\hat{\Delta}(\alpha, \gamma) = (\hat{F} + \gamma I)^{-\alpha/2} (-\hat{g}_N^{\text{hard}}(0)),$$

where α controls inverse strength and γ damps poorly estimated flat directions (Tikhonov & Arsenin, 1977; Engl et al., 1996). We estimate the full $d_h \times d_h$ Fisher at the unsteered model using 30k clean carrier contexts and four resampled targets per context. Appendix A.4 describes the Rademacher estimator for F .

Figure 2 compares the uncorrected mean gradient with the best damped Fisher-inverted estimate. Across all five traits, inversion produces a large alignment gain. This cross-trait consistency supports a shared geometric explanation in which the carrier task reweights the components of Δ_T .

The diagnostic deliberately uses access to the known teacher vector to select the optimal (α, γ) pair separately for each trait. It therefore tests the predicted Fisher form of the distortion rather than providing a teacher-free estimator. The consistent improvement shows that the initial gradient is not merely weak or noisy: it is transformed by carrier curvature. Explicit correction nevertheless remains below ordinary iterative recovery. The next section explains this gap by showing that iterative optimization progressively inverts the same transformation.

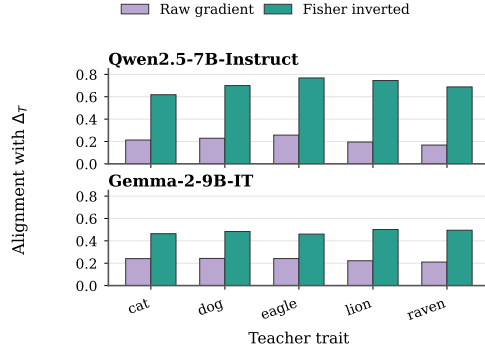


Figure 2: Fisher inversion improves single-step alignment across models and traits. Hyperparameters are selected against Δ_T .

5 ITERATION PROGRESSIVELY INVERTS THE CARRIER TRANSFORMATION

5.1 ONE DYNAMICS CONTAINS HARD AND SOFT SUPERVISION

Extending the local model in Equation 5 along the bounded trajectory gives

$$\nabla_{\Delta_S} \mathcal{L}_N^{\text{hard}}(\Delta_S) \approx F(\Delta_S - \Delta_T) + \bar{\xi}_N, \quad (6)$$

where $\bar{\xi}_N$ is the fixed hard-label sampling error characterized in Section 4.1. Soft supervision is the noiseless special case $\bar{\xi}_N = 0$. For this soft objective, $\Delta_S = \Delta_T$ matches the teacher distribution

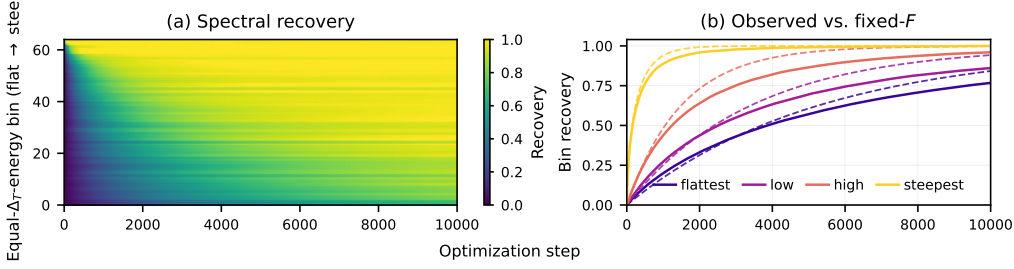


Figure 3: **Iteration progressively restores flatter directions.** (a) Recovery of 64 equal- Δ_T -energy Fisher bins under soft-KL optimization; curvature increases from bottom to top. (b) Observed recovery for representative coarse bins (solid) and the fixed- F prediction $\phi_k(\lambda)$ (dashed).

on every observed prefix and yields zero KL exactly. Thus curvature changes the optimization path without changing the noiseless target.

Approximating F and $\bar{\xi}_N$ as constant along the local trajectory, gradient descent with learning rate η obeys (Landweber, 1951; Engl et al., 1996)

$$\Delta_S^{(k+1)} = \Delta_S^{(k)} - \eta \left[F(\Delta_S^{(k)} - \Delta_T) + \bar{\xi}_N \right]. \quad (7)$$

Starting from $\Delta_S^{(0)} = 0$, its solution is

$$\Delta_S^{(k)} = [I - (I - \eta F)^k] \Delta_T - P_k \bar{\xi}_N, \quad P_k := \eta \sum_{j=0}^{k-1} (I - \eta F)^j. \quad (8)$$

The derivation is given in Appendix A.3. Setting $\bar{\xi}_N = 0$ gives the soft-supervision trajectory

$$\Delta_S^{(k)} = [I - (I - \eta F)^k] \Delta_T. \quad (9)$$

Writing $F = \sum_i \lambda_i u_i u_i^\top$ and $\Delta_T = \sum_i \delta_i u_i$ makes this noiseless trajectory explicit:

$$\Delta_S^{(k)} = \sum_i \underbrace{[1 - (1 - \eta \lambda_i)^k]}_{\phi_k(\lambda_i)} \delta_i u_i. \quad (10)$$

The eigenvalue λ_i determines the convergence speed of direction u_i : directions with larger eigenvalues approach their teacher components more quickly, whereas directions with smaller eigenvalues require more steps (Yao et al., 2007; Bauer et al., 2007; Lo Gerfo et al., 2008; Advani et al., 2020). $\phi_k(\lambda_i) \rightarrow 1$ as $k \rightarrow \infty$ for every $\lambda_i > 0$. Thus, although different directions converge at different rates, every component is eventually restored.

We test this prediction in the noise-free setting by training on soft labels from **only 500** Qwen carrier sequences and projecting all 10K iterates into the Fisher eigenbasis. Figure 3 groups directions into 64 contiguous bins with approximately equal teacher-vector energy. Recovery advances from steep to flat bins, as predicted by Equation 10. At step 10,000, recovery is 0.77 in the flattest coarse bin and 0.99 in the steepest. The observed vector has cosine 0.969 with the trajectory predicted by the fixed- F model, showing that the model captures the ordering and most of the trajectory while mildly overestimating the speed of flat-direction recovery.

5.2 ITERATION IMPLEMENTS A PROGRESSIVELY LESS DAMPED INVERSE

The same dynamics can be written as an operation on the initial learning signal:

$$\Delta_S^{(k)} = P_k(F\Delta_T), \quad (11)$$

Along an eigenvector of F with eigenvalue $\lambda > 0$, the operator P_k has scalar response

$$p_k(\lambda) = \frac{1 - (1 - \eta \lambda)^k}{\lambda}. \quad (12)$$

This expression connects the two empirical endpoints. Let K denote the number of optimization steps performed after the initial gradient update. At $K = 0$ (equivalently, $k = 1$ above), $P_1 = \eta I$ and no inverse correction has yet occurred. The recovered direction is therefore $F\Delta_T$, explaining the shared low single-step cosine similarity in Section 3. As $K \rightarrow \infty$, $p_k(\lambda) \rightarrow 1/\lambda$ for every direction. The carrier transformation is then fully inverted and $P_k F\Delta_T \rightarrow \Delta_T$, explaining the near-perfect recovery obtained by iterative KL optimization. Intermediate values of K interpolate between these endpoints by correcting progressively flatter directions (Ali et al., 2019).

The remaining question is, the soft and hard supervision gradient converge as N grows, then why hard supervision does not follow the same trajectory as the soft one. We address this next by analyzing how discretization noise is amplified as optimization attempts to invert progressively flatter directions.

6 DATA SCALE CONTROLS THE DEPTH OF HARD-LABEL INVERSION

Section 5.2 shows that optimization progressively removes the carrier transformation, reaching flatter components only at later steps. Under hard supervision, however, each additional stage of inversion acts not only on the teacher signal but also on the fixed sampling error derived in Section 4.1. Recovery therefore has a data-dependent optimal depth: early updates reduce carrier-induced bias, whereas sufficiently late updates amplify sampled-label error faster than they recover the remaining signal. We show that increasing the number of independent carriers postpones this crossover, allowing the student to follow the noiseless trajectory longer and recover progressively flatter components.

6.1 MORE DATA PERMITS DEEPER INVERSION BEFORE NOISE DOMINATES

The hard and soft mean gradients in Section 3 are nearly collinear, yet their iterative endpoints differ sharply. The reason is that optimization simultaneously removes carrier-induced bias and amplifies the small hard-label error fixed by the sampled dataset. Combining $\text{Cov}(\bar{\xi}_N) \approx F/N$ from Section 4.1 with Equation 8, and writing $F = \sum_i \lambda_i u_i u_i^\top$, gives

$$\hat{\delta}_i^{(k)} = \phi_k(\lambda_i)\delta_i - p_k(\lambda_i)\xi_i, \quad p_k(\lambda) = \frac{1 - (1 - \eta\lambda)^k}{\lambda}, \quad (13)$$

where $\delta_i = \langle \Delta_T, u_i \rangle$ and $\xi_i = \langle \bar{\xi}_N, u_i \rangle$. Since $\text{Var}(\xi_i) \approx \lambda_i/N$ and the noise is mean zero, the expected component-wise error separates into

$$\mathbb{E}[(\hat{\delta}_i^{(k)} - \delta_i)^2] \approx \underbrace{[1 - \phi_k(\lambda_i)]^2 \delta_i^2}_{\text{remaining inversion bias}} + \underbrace{\frac{\lambda_i}{N} p_k(\lambda_i)^2}_{\text{amplified label noise}}. \quad (14)$$

Increasing k decreases the first term but increases the second toward $1/(N\lambda_i)$. Hard-label training therefore has a noise-limited optimal depth k^* : beyond it, further inversion increases total error. Increasing N lowers only the variance term, so it should postpone k^* and permit optimization to recover progressively slower, flatter components before noise dominates (Raskutti et al., 2014).

We test this prediction using matched hard and soft Qwen cat-carrier trajectories with $M \in \{500, 1000, 2000, 4000, 8000, 16000\}$ independent carrier sequences. Their nearly constant mean completion length makes the supervised token count N proportional to M . At each logged step we project into 64 equal-teacher-energy bins of the clean-carrier Fisher basis and average the normalized hard error

$$R^H(k) = \frac{1}{64} \sum_{b=1}^{64} \frac{\|P_b(\Delta_S^H(k) - \Delta_T)\|_2^2}{\|P_b\Delta_T\|_2^2}, \quad (15)$$

with the soft error defined analogously. The model, teacher vector, optimizer, and 1,000-step horizon are otherwise fixed.

The left panel of Figure 4 exhibits precisely this tradeoff. For $N = 500$, hard error reaches its minimum after only 100 updates and then rises to 2.756 by step 1,000, even as the soft reference keeps improving. As M doubles, k^* moves systematically from 100 to 190, 330, 560, and 880

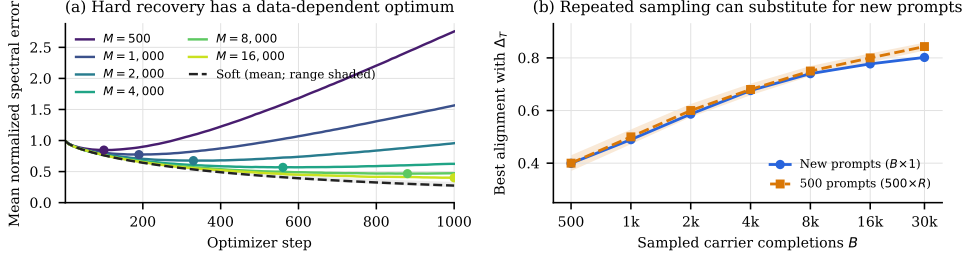


Figure 4: **Left:** hard-label spectral error initially falls, then rises as amplified noise overtakes the remaining inversion bias; dots mark each minimum. **Right:** best alignment at a matched completion budget B , or repeatedly sampling generation from a fixed pool of 500 prompts for $B/500$ times.

updates. Larger datasets delay the point at which inverse-amplified sampling error overtakes the remaining bias, allowing the student to follow the noiseless trajectory longer and recover flatter components. The right panel is designed to understand the effectiveness of independent data. At a matched completion budget B , it compares expanding the prompt pool with repeatedly sampling from a fixed pool of 500 prompts. We found that 500 independent samples is sufficient for recovery; the effectiveness of adding more independent samples is equivalent for reducing hard label noise via sampling more responses from the small prompt pool.

6.2 TRAIT SPECTRAL LOCATION IS ASSOCIATED WITH RECOVERY DIFFICULTY

The factor $1/(N\lambda_i)$ predicts that recovery should be harder when more of the teacher direction lies at low carrier curvature. We test this prediction across Qwen2.5-7B and Gemma-2-9B using the same five animal traits. For each model, we project Δ_T into its clean-carrier Fisher eigenbasis and plot the energy $\langle \Delta_T, u_i \rangle^2 / \|\Delta_T\|^2$ against normalized curvature $\lambda_i/\bar{\lambda}$ (Figure 5) (Hansen, 1998). Across all five traits, the main mass of Gemma’s distribution is shifted toward lower normalized curvature. This shift is consistent with Gemma’s lower recovery, because more of its teacher direction must be inferred through slow, noise-sensitive components.

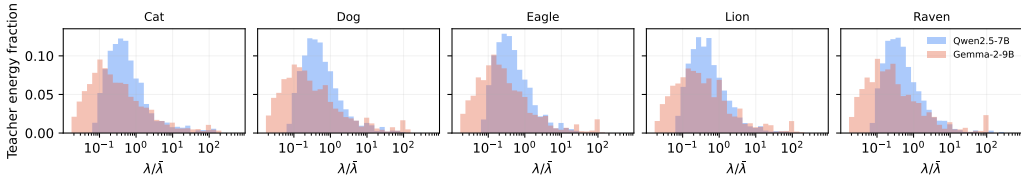


Figure 5: **Gemma trait energy lies at lower normalized curvature.** Teacher-vector energy across common bins of $\lambda/\bar{\lambda}$ for five traits; each spectrum is normalized by its model-specific mean.

7 CONCLUSION

We asked whether subliminal learning requires large carrier datasets because the trait is insufficiently represented or because sampled targets introduce error. Controlled experiments show that hard and soft supervision produce nearly identical but distorted initial gradients, yet soft supervision recovers the teacher direction from hundreds of carriers while hard recovery remains limited. Our noisy-inverse model explains why: carrier Fisher geometry transforms the teacher direction, iteration reverses this transformation, and the same inverse amplifies fixed hard-label error, especially in weakly observed directions. Large datasets therefore make deeper inversion reliable rather than merely revealing an absent trait. This unifies the shared single-step ceiling, divergent iterative endpoints, and the advantage of new independent data over repeated passes on fixed samples.

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A DERIVATIONS AND FISHER ESTIMATION

A.1 HARD-LABEL NOISE COVARIANCE

For a fixed prefix s_t and $y_t \sim p_{T,t}$, the one-hot target satisfies

$$\mathbb{E}[e_{y_t} | s_t] = p_{T,t}, \quad \mathbb{E}[e_{y_t} e_{y_t}^\top | s_t] = \text{Diag}(p_{T,t}).$$

The second identity follows because $e_{y_t} e_{y_t}^\top$ contains one nonzero diagonal entry at the sampled class. Therefore

$$\begin{aligned} \text{Cov}(e_{y_t} | s_t) &= \mathbb{E}[e_{y_t} e_{y_t}^\top | s_t] - \mathbb{E}[e_{y_t} | s_t] \mathbb{E}[e_{y_t} | s_t]^\top \\ &= \text{Diag}(p_{T,t}) - p_{T,t} p_{T,t}^\top =: \Lambda_{T,t}. \end{aligned}$$

Since $\xi_t = J_t^\top (p_{T,t} - e_{y_t})$,

$$\mathbb{E}[\xi_t | s_t] = 0, \quad \text{Cov}(\xi_t | s_t) = J_t^\top \Lambda_{T,t} J_t =: F_{T,t}.$$

For N independent supervised tokens,

$$\begin{aligned} \text{Cov}(\bar{\xi}_N) &= \frac{1}{N^2} \sum_{t=1}^N \mathbb{E}_{s_t}[F_{T,t}] \\ &\approx \frac{F}{N}. \end{aligned}$$

The final approximation uses the local regime $p_{T,t} \approx p_{0,t}$, under which the teacher-position Fisher averages to the base-model carrier Fisher. Within-sequence correlations can change the finite-data covariance.

For $Fu_i = \lambda_i u_i$, $\|u_i\|_2 = 1$, and $\xi_i = u_i^\top \bar{\xi}_N$,

$$\begin{aligned} \text{Var}(\xi_i) &= u_i^\top \text{Cov}(\bar{\xi}_N) u_i \\ &\approx \frac{1}{N} u_i^\top F u_i = \frac{\lambda_i}{N}. \end{aligned}$$

A.2 SCORE IDENTITY AND DISTRIBUTION MISMATCH

For a fixed prefix, J_t does not depend on the sampled target. If $y \sim p_0(\cdot | s_t)$, then

$$\begin{aligned} \mathbb{E}_y[g_t(y) g_t(y)^\top] &= J_t^\top \mathbb{E}_y[(p_{0,t} - e_y)(p_{0,t} - e_y)^\top] J_t \\ &= J_t^\top [\text{Diag}(p_{0,t}) - p_{0,t} p_{0,t}^\top] J_t \\ &= J_t^\top \Lambda_t J_t = F_t. \end{aligned}$$

If targets instead come from the teacher while scores are evaluated at the base model,

$$\mathbb{E}_{y \sim p_{T,t}}[(p_{0,t} - e_y)(p_{0,t} - e_y)^\top] = \Lambda_{T,t} + (p_{T,t} - p_{0,t})(p_{T,t} - p_{0,t})^\top.$$

The second term contains teacher signal rather than carrier curvature. This is why the correction experiment estimates F on clean base-model carriers.

A.3 FINITE-STEP SOLUTION

Under the constant- F local model, gradient descent obeys

$$\Delta_S^{(k+1)} = (I - \eta F)\Delta_S^{(k)} + \eta(F\Delta_T - \bar{\xi}_N).$$

Starting from zero and unrolling gives

$$\Delta_S^{(k)} = \eta \sum_{j=0}^{k-1} (I - \eta F)^j (F\Delta_T - \bar{\xi}_N) = P_k(F\Delta_T - \bar{\xi}_N),$$

which is Equation 8. The geometric-series identity $P_k F = I - (I - \eta F)^k$ yields the soft trajectory in Equation 9.

A.4 RADEMACHER ESTIMATION ON CLEAN CARRIERS

Let M be the number of clean carrier contexts and $\mathcal{I}_m$ the completion-token indices in context m . For each of R draws, sample

$$\tilde{y}_t^{(r)} \sim p_0(\cdot | s_t), \quad \rho_t^{(r)} \sim \text{Unif}\{-1, +1\},$$

and compute in one backward pass

$$q_m^{(r)} = \frac{1}{\sqrt{|\mathcal{I}_m|}} \sum_{t \in \mathcal{I}_m} \rho_t^{(r)} g_t(\tilde{y}_t^{(r)}).$$

Our sequence-equal estimate is

$$\hat{F} = \frac{1}{MR} \sum_{m=1}^M \sum_{r=1}^R q_m^{(r)} q_m^{(r)\top}.$$

Because independent Rademacher signs satisfy $\mathbb{E}[\rho_t \rho_{t'}] = 0$ for $t \neq t'$,

$$\mathbb{E}_\rho[q_m^{(r)} q_m^{(r)\top}] = \frac{1}{|\mathcal{I}_m|} \sum_{t \in \mathcal{I}_m} g_t(\tilde{y}_t^{(r)}) g_t(\tilde{y}_t^{(r)})^\top.$$

Taking expectation over resampled targets and applying the score identity gives the average token Fisher. We use $M = 30,000$ and $R = 4$. Resampling from p_0 and using clean carrier contexts prevents the teacher signal from contaminating the curvature estimate.

B RELATED WORK

Subliminal learning. Knowledge distillation conventionally transfers a teacher’s predictive behavior by matching its outputs (Hinton et al., 2015). Subliminal learning shows that model-generated data can additionally transmit behavioral traits that are not semantically expressed in those data (Cloud et al., 2026). Subsequent work has studied when this transfer occurs across architectures, carrier tasks, and intervention parameterizations (Schrodi et al., 2025; Blank et al., 2026). These studies establish the generality of hidden trait transfer, whereas we focus on the optimization geometry that determines how accurately a known trait direction can be recovered.

Steering-vector accounts. Recent work connects subliminal transfer to the distillation of latent steering directions (Blank et al., 2026) and introduces subliminal steering as a controlled way to encode stronger, directly measurable hidden signals (Morgulis & Hewitt, 2026). We adopt this controlled setting because it exposes both the teacher direction and the recovered student direction. Our analysis complements these accounts by showing that the carrier task first transforms the teacher direction through its Fisher geometry and that iterative optimization acts as an increasingly strong inverse of that transformation.

Data scale and signal localization. Empirically, increasing the number of independently generated distillation examples makes latent teacher traits more recoverable (Dong et al., 2026). Related evidence emphasizes that independent carrier diversity is more useful than repeatedly training on a fixed dataset and links transfer to token positions at which teacher predictions diverge (Feng, 2026). These observations explain where trait-dependent signals can appear, but do not characterize why recovery continues to depend on large independent datasets after the mean learning direction has stabilized. Our noisy-inverse account separates these roles: carrier geometry determines the systematic transformation of the trait, while independent hard targets control the residual discretization error that is amplified as optimization inverts that transformation.

Inverse problems and iterative regularization. The locally linearized recovery dynamics are a finite-dimensional instance of Landweber iteration (Landweber, 1951): the iteration count acts as a regularization parameter, and each Fisher eigendirection is governed by a spectral filter. This perspective is classical in inverse problems (Tikhonov & Arsenin, 1977; Engl et al., 1996; Hansen, 1998), including the use of Tikhonov-style damping and early stopping to suppress unstable low-curvature modes. Statistical inverse-problem theory further characterizes the corresponding bias-variance tradeoff when the observation error is random and decreases with sample size (Bissantz et al., 2007; Cavalier, 2008). Our plots of trait energy across the Fisher spectrum are also related to spectral source and discrete Picard conditions, which describe how the target coefficients are distributed relative to an ill-posed operator’s spectrum (Hansen, 1998).

Spectral filtering and early stopping in learning. Gradient descent has likewise been studied as a spectral regularization method in learning theory (Yao et al., 2007; Bauer et al., 2007; Lo Gerfo et al., 2008). This literature establishes consistency and stopping rules for iterative regularization (Raskutti et al., 2014), relates gradient-flow paths to ridge regularization (Ali et al., 2019), and analyzes mode-dependent learning and overtraining in high-dimensional models (Advani et al., 2020). Our contribution is therefore not the general observation that gradient descent implements a spectral filter. We identify the carrier Fisher as the relevant forward operator for subliminal steering, derive the hard-label forcing term and its sample-size dependence, and test the resulting mode-wise and data-scaling predictions in an auditable parameter-recovery setting.

Fisher geometry and distributional supervision. Fisher preconditioning is central to natural-gradient methods (Amari, 1998); importantly, the model Fisher used here is distinct from the observed-label outer product often called the empirical Fisher (Kunstner et al., 2019). Prior statistical accounts of knowledge distillation show that replacing sampled hard labels with teacher probabilities can reduce the variance of the student objective (Menon et al., 2021). In our setting, this variance reduction has a specific inverse-problem consequence: distributional supervision retains the carrier-transformed signal while removing the conditional token-sampling noise that would otherwise be amplified along low-curvature directions.

C HYPERPARAMETERS

Table 1 reports the model and data configuration. All pretrained parameters are frozen; only one zero-initialized residual vector is optimized and shared across the listed layers. Unless stated otherwise, we use seed 42, BF16 precision, teacher scale $\alpha = 1$, and temperature 1.

D LIMITATIONS AND FUTURE WORK

Our analysis is local. It linearizes the model around the unsteered state and treats the carrier Fisher matrix F as approximately constant along the bounded recovery trajectory. This approximation captures the observed curvature-dependent ordering of recovery, but changes in the model Jacobian and Fisher geometry can account for deviations from the fixed- F dynamics.

We use subliminal steering as an experimental platform because it turns hidden trait transfer into an auditable parameter-recovery problem: both the implanted teacher direction and the recovered student direction are known, so recovery can be measured directly rather than inferred only from behavior. Representing the teacher trait by a single low-dimensional steering vector also makes

Configuration	Qwen2.5-7B-Instruct	Gemma-2-9B-IT
Hidden width / transformer layers	3584 / 28	3584 / 42
Intervened layers	[2, 26)	[2, 40)
Teacher norm bound	0.5	1.5
Traits	cat, dog, eagle, lion, raven	
Teacher training examples	50	
Teacher optimizer / learning rate	Adam / 10^{-3}	Adam / 10^{-3}
Teacher optimization steps / batch size	200 / 10	200 / 10
Carrier temperature / top- k	1.0 / full vocabulary (no truncation)	
Recovery objectives	hard NLL and soft KL	
Recovery optimizer / learning rate	SGD / 10^{-2}	SGD / 0.5

Table 1: Model, intervention, teacher-extraction, carrier-generation, and student-recovery hyperparameters. The same recovery optimizer and learning rate are used for hard and soft supervision.

an explicit Fisher analysis computationally feasible. This control does not cover the full range of subliminal learning. Traits induced by system prompts and students trained through LoRA or full-parameter fine-tuning may involve larger, higher-dimensional, and more nonlinear changes than the bounded steering trajectories studied here. Whether the same noise-dominated inverse mechanism accounts for their data requirements therefore remains to be tested across broader teacher constructions, student parameterizations, and carrier tasks; we leave those further validations for future work.

AI USE STATEMENT

Generative AI tools were used to assist with code implementation and debugging, experimental analysis, figure and table preparation, and language editing and organization of the manuscript. The authors reviewed and verified all AI-assisted outputs, independently validated the reported experimental results, and take full responsibility for the scientific claims and final content.

REPRODUCIBILITY STATEMENT

Code is available at <https://github.com/luoyuchenmlcv/subliminal-data>.